KATHIR V

Junior Machine Learning Engineer | Healthcare AI

Udumalaippettai, Tamil Nadu | Open to Chennai, Coimbatore, Bengaluru | +91 8807662912 | kathir4605@gmail.com | linkedin.com/in/kathir-v | github.com/Kathir1730

PROFESSIONAL SUMMARY

Fresher ML engineer (B.Tech Bioinformatics, 2026) with project experience in healthcare audio, medical imaging and protein sequence data. Built a lung-sound classifier reaching 91.59% accuracy across 14 conditions, and trained a Vision Transformer from scratch to 99.18% test accuracy. Skilled in Python, PyTorch, TensorFlow, scikit-learn and feature engineering.

TECHNICAL SKILLS

Languages: Python, SQL (basic), R

Machine Learning: scikit-learn, XGBoost, Random Forest, Extra Trees, SVM, KNN, Ensemble Learning, Hyperparameter Tuning, Feature Engineering, SMOTE+ENN, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras, Vision Transformer (ViT), MobileNetV2, MLP, Score-CAM

Data & Tools: NumPy, Pandas, Librosa, Jupyter Notebook, GitHub, Excel, Biopython

PROJECTS

Safe Breath: Lung Sound Classification (Healthcare AI) | Python, Librosa, scikit-learn, TensorFlow, XGBoost | Google Play: play.google.com/store/apps/details?id=com.gs.safebreath

- Classified 14 respiratory-condition categories at 91.59% accuracy (XGBoost) and 91.24% (MLP), by training and tuning MLP, Random Forest, Extra Trees, XGBoost and an MLP + Random Forest ensemble on 3,530 recordings (7,821 augmented samples).
- Built a 146-dimensional acoustic feature set (MFCC, spectral features, ZCR, RMS energy), after wavelet denoising, silence removal, normalization and 5-second segmentation.
- Addressed class imbalance using data augmentation and SMOTE+ENN. Contributed UI suggestions for the Safe Breath Android app that uses this model.

Blood Cell Classification using Vision Transformer | Python, PyTorch, ViT, Score-CAM

- Achieved 99.18% test accuracy on 3 classes (RBC, WBC, platelets), by training ViT-Base (Patch16-224) from scratch for 50 epochs on 4,886 cropped BCCD cell images with augmentation and noise addition (70:15:15 split).
- Showed which cell regions drove predictions, using Score-CAM visualizations for model interpretability.

Physicochemical-Enhanced Embeddings for Protein Family Classification (Team Project) | Python, Biopython, scikit-learn

- Raised accuracy from 16.95% (standard 3-mer baseline) to 64.63%, while cutting features from 1,000 to 261 (74% fewer), by encoding sequences as 3-mers of hydropathy-grouped amino acids on 11,620 UniProt/Swiss-Prot sequences across 10 families.
- Implemented and compared Random Forest, SVM, KNN and Extra Trees, evaluated on an 80:20 stratified split using accuracy, F1-score and confusion matrices.
- Created performance visualizations and optimized the modelling workflow.

Dog Breed Image Classification | Python, TensorFlow/Keras, MobileNetV2

- Built a 120-breed image classifier, by fine-tuning a MobileNetV2 architecture in TensorFlow/Keras.

EDUCATION

B.Tech, Bioinformatics | SASTRA Deemed University, Thanjavur | Aug 2022 – May 2026 | CGPA: 7.46/10

CERTIFICATIONS

- AI & Machine Learning and Data Science – Udemy, 2024
- Generative AI & Prompt Engineering – HCLTech, 2025